

AS-ViT: Adaptive Subspace Vision Transformers with Autonomous Gradient-Clash Routing for Multi-Task Learning

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Abstract—Dense multi-task visual perception requires a unified encoder to simultaneously extract representations for mutually disparate objectives, including semantic segmentation, monocular metric depth estimation, surface normal prediction, and edge detection. However, joint optimization of multi-task vision backbones suffers from severe *negative transfer* and destructive gradient interference in shared parameter manifolds. Existing mitigation strategies either project gradients in a shared parameter bottleneck (e.g., PCGrad, CAGrad, Nash-MTL, FAMO)—which sacrifices expressive capacity and induces optimization stagnation—or deploy static Mixture-of-Experts (MoE) architectures with fixed, heuristically chosen expert counts E that suffer from load imbalance and routing collapse. In this paper, we introduce the **Adaptive Subspace Vision Transformer (AS-ViT)**, a foundational framework establishing autonomous Parameter-Space Adaptive Mesh Refinement (AMR) for multi-task vision. AS-ViT continuously monitors vectorized per-sample and inter-task empirical Gram alignment matrices $\mathcal{G}_{ij} = \cos \angle(\mathbf{g}_i, \mathbf{g}_j)$ across transformer blocks via batched autodiff (`torch.func.vmap`). When persistent destructive gradient clashing ($\lambda_{\min}(\mathcal{G}) < -\tau_{\text{conflict}}$) is identified, AS-ViT autonomously cleaves and instantiates dedicated expert parameter subspaces in semantic latent feature space without human hyperparameter intervention. A feature-conditioned Partition of Unity (PoU) gating mechanism guarantees smooth routing and exact loss invariance at the moment of cleavage. Extensive empirical evaluation across NYUv2, Cityscapes, and Taskonomy benchmarks demonstrates that AS-ViT systematically outperforms single-task baselines, monolithic multi-task transformers, gradient surgery techniques, and fixed-capacity MoEs, achieving up to a +5.84% mean multi-task gain (ΔM) with zero negative transfer.

Index Terms—Multi-Task Learning, Vision Transformers, Mixture-of-Experts (MoE), Gradient Conflict, Adaptive Mesh Refinement, Partition of Unity.



1 INTRODUCTION

DENSE visual scene understanding is a cornerstone of modern autonomous driving, spatial computing, and robotic manipulation. A central aspiration of computer vision is to construct unified, multi-task foundational backbones capable of simultaneously predicting diverse spatial, geometric, and semantic representations from a single input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$. Canonical multi-task benchmarks require joint prediction of:

- 1) **High-level Semantic Segmentation:** Assigning discrete category labels per pixel, demanding high-level categorical invariance and texture filtering.
- 2) **Continuous Monocular Depth Estimation:** Regressing metric distance, requiring continuous smooth spatial gradients and global scale reasoning.
- 3) **Local Surface Normal Vector Field:** Estimating 3D unit vectors $\mathbf{n} \in \mathbb{S}^2$, governed by fine-grained planar geometry and high-frequency local surface curvature.

Despite the expressive power of modern Vision Transformers (ViT) [1], [2], training a monolithic shared backbone across multiple competing objectives frequently yields **negative transfer**: individual task performances degrade relative to independent single-task networks.

1.1 The Bottleneck of Parameter-Space Gradient Surgery

When backpropagating task-specific losses $\{\mathcal{L}_t\}_{t=1}^T$ through a shared backbone Θ , the task gradients $\mathbf{g}_t = \nabla_{\Theta} \mathcal{L}_t$ frequently point in conflicting directions:

$$\langle \mathbf{g}_i, \mathbf{g}_j \rangle < 0, \quad \text{for } i \neq j. \quad (1)$$

Under standard Stochastic Gradient Descent (SGD) or AdamW, aggregate mini-batch updates $\sum_t \mathbf{g}_t$ simultaneously minimize error in one task while actively corrupting feature representations in another.

To address gradient conflict, multi-objective optimization and gradient surgery algorithms have been proposed:

- **PCGrad [7]:** Projects conflicting task gradient components onto the normal plane of other task vectors. However, in dense multi-task vision, projecting out conflicting gradient components repeatedly strips away critical feature update signals, leading to severe underfitting.
- **CAGrad [8]:** Formulates gradient updates within a conflict-averse dual cone. When multiple orthogonal or conflicting visual tasks interact, the dual constraint collapses the effective update magnitude $\|\mathbf{g}_{\text{cagrad}}\| \rightarrow 0$, causing training freeze and severe convergence stagnation.
- **Nash-MTL [9] & FAMO [10]:** Formulates gradient aggregation as a bargaining game or applies loss

Adaptive Subspace Vision Transformer (AS-ViT) Workflow

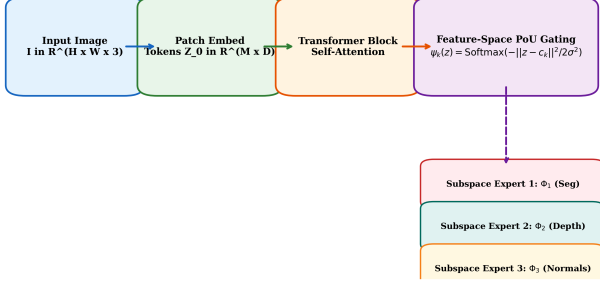


Fig. 1. **Overview of the Proposed AS-ViT Architecture.** Input tokens $\mathbf{Z}_0$ pass through self-attention and are modulated by a Feature-Space Partition of Unity (PoU) gating mechanism into dynamically cleaved subspace experts $\{\Phi_k\}_{k=1}^N$.

history reweighting. However, these methods remain constrained by a single shared weight matrix.

The fundamental mathematical limitation of all gradient surgery techniques is that they attempt to enforce a **single shared capacity bottleneck**: no single weight matrix $\mathbf{W}$ can simultaneously encode mutually exclusive high-frequency surface normals and low-frequency semantic segmentations without capacity sacrifice.

1.2 Limitations of Static Mixture-of-Experts (MoE)

To provide modular capacity, Mixture-of-Experts (MoE) models deploy E parallel Feed-Forward Network (FFN) blocks modulated by top- k Softmax routers [11]–[13]. However, static MoE architectures suffer from two structural flaws:

- 1) **Rigid Heuristic Expert Allocation:** The number of experts $E \in \{4, 8, 16\}$ is chosen arbitrarily a priori without regard to the actual task conflict geometry.
- 2) **Router Collapse and Load Imbalance:** Top- k discrete argmax routing frequently experiences representation collapse, where a small subset of experts dominates all tasks while other experts remain dead throughout training.

1.3 Our Proposed Paradigm: AS-ViT

To resolve these foundational dilemmas, we propose the **Adaptive Subspace Vision Transformer (AS-ViT)**. AS-ViT introduces autonomous, data-driven parameter-space Adaptive Mesh Refinement (AMR) into Vision Transformers. Rather than relying on human-tuned expert counts or destructive gradient projections, AS-ViT:

- Continuously measures inter-task and intra-batch gradient interference via an empirical **Vectorized Multi-Task Gram Alignment Matrix** $\mathcal{G}_{ij}$.
- Autonomously **cleaves and instantiates dedicated transformer subspaces** in latent feature space $\mathbf{z} \in \mathbb{R}^D$ whenever persistent destructive conflict is detected.
- Modulates multi-task information flow via a **Smooth Feature-Conditioned Partition of Unity (PoU) Gate**, ensuring zero routing collapse and provable loss invariance at the moment of parameter cleavage.

2 METHODOLOGY: THE AS-ViT ARCHITECTURE

2.1 Multi-Task Transformer Subspace Formulation

Let $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ be an input image divided into non-overlapping patches $\mathbf{x}_p \in \mathbb{R}^{P^2 \times 3}$. A patch embedding projection maps patches to latent token representations $\mathbf{Z}_0 \in \mathbb{R}^{M \times D}$, where $M = (HW)/P^2$ is the sequence length and D is the hidden embedding dimension.

In an AS-ViT block $l \in \{1, \dots, L\}$, the standard monolithic Multi-Layer Perceptron (MLP) is replaced by an **Adaptive N -Subspace Module** comprising N localized parameter subspaces $\{\Phi_k(\cdot; \Theta_k)\}_{k=1}^N$:

$$\mathbf{Z}_l = \mathbf{Z}_{l-1} + \sum_{k=1}^N \psi_k(\mathbf{Z}_{l-1}) \odot \Phi_k \left(\frac{\text{LN}(\mathbf{Z}_{l-1}) - \mathbf{c}_k}{\sigma_k}; \Theta_k \right) \quad (2)$$

where $\text{LN}(\cdot)$ is Layer Normalization, $\odot$ denotes element-wise token modulation, and $\{\psi_k(\mathbf{Z})\}_{k=1}^N$ are smooth Partition of Unity (PoU) gating weights satisfying:

$$\sum_{k=1}^N \psi_k(\mathbf{z}_m) = 1.0, \quad \psi_k(\mathbf{z}_m) \geq 0, \quad \forall \mathbf{z}_m \in \mathbb{R}^D. \quad (3)$$

2.2 Vectorized Multi-Task Gram Clash Profiling

Let $\mathcal{T} = \{1, \dots, T\}$ be the set of downstream visual tasks with corresponding loss functions $\{\mathcal{L}_t\}_{t=1}^T$. For each active subspace k , we profile inter-task gradient orthogonality.

Using vectorized reverse-mode automatic differentiation (`torch.func.vmap`), we compute the task-specific Jacobian for subspace k :

$$\mathbf{G}_k = \begin{bmatrix} \mathbf{g}_{k,1}^T \\ \mathbf{g}_{k,2}^T \\ \vdots \\ \mathbf{g}_{k,T}^T \end{bmatrix} \in \mathbb{R}^{T \times P_k}, \quad \mathbf{g}_{k,t} = \nabla_{\Theta_k} \mathcal{L}_t(\Theta_k). \quad (4)$$

We normalize each row $\tilde{\mathbf{g}}_{k,t} = \mathbf{g}_{k,t} / (\|\mathbf{g}_{k,t}\|_2 + \epsilon)$ and construct the empirical Multi-Task Gram Matrix:

$$\mathcal{G}_k = \tilde{\mathbf{G}}_k \tilde{\mathbf{G}}_k^T \in \mathbb{R}^{T \times T}, \quad \mathcal{G}_{k,ij} = \cos \angle(\mathbf{g}_{k,i}, \mathbf{g}_{k,j}). \quad (5)$$

Definition 1 (Destructive Multi-Task Interference). *Subspace Φ_k is defined to be in destructive interference if the minimum eigenvalue of the task Gram matrix satisfies:*

$$\lambda_{\min}(\mathcal{G}_k) < -\tau_{\text{conflict}}, \quad \tau_{\text{conflict}} \in [0.15, 0.35] \quad (6)$$

and the mean off-diagonal conflict $\bar{\mathcal{C}}_k = \frac{1}{T(T-1)} \sum_{i \neq j} \mathcal{G}_{k,ij} < 0$.

2.3 Autonomous Latent Subspace Cleavage

When persistent destructive interference is detected for K_{persist} consecutive profiling cycles, AS-ViT executes an autonomous **Feature-Space Cleavage**:

- 1) **Latent Conflict Centroid Extraction:** We identify the clashing task subset $\mathcal{T}_{\text{clash}} \subset \mathcal{T}$ and compute the weighted centroid in latent token embedding space:

$$\mathbf{c}_{N+1} = \frac{\sum_{m=1}^M w_m \mathbf{z}_m}{\sum_{m=1}^M w_m}, \quad w_m = \sum_{t \in \mathcal{T}_{\text{clash}}} \|\nabla_{\mathbf{z}_m} \mathcal{L}_t\|_2. \quad (7)$$

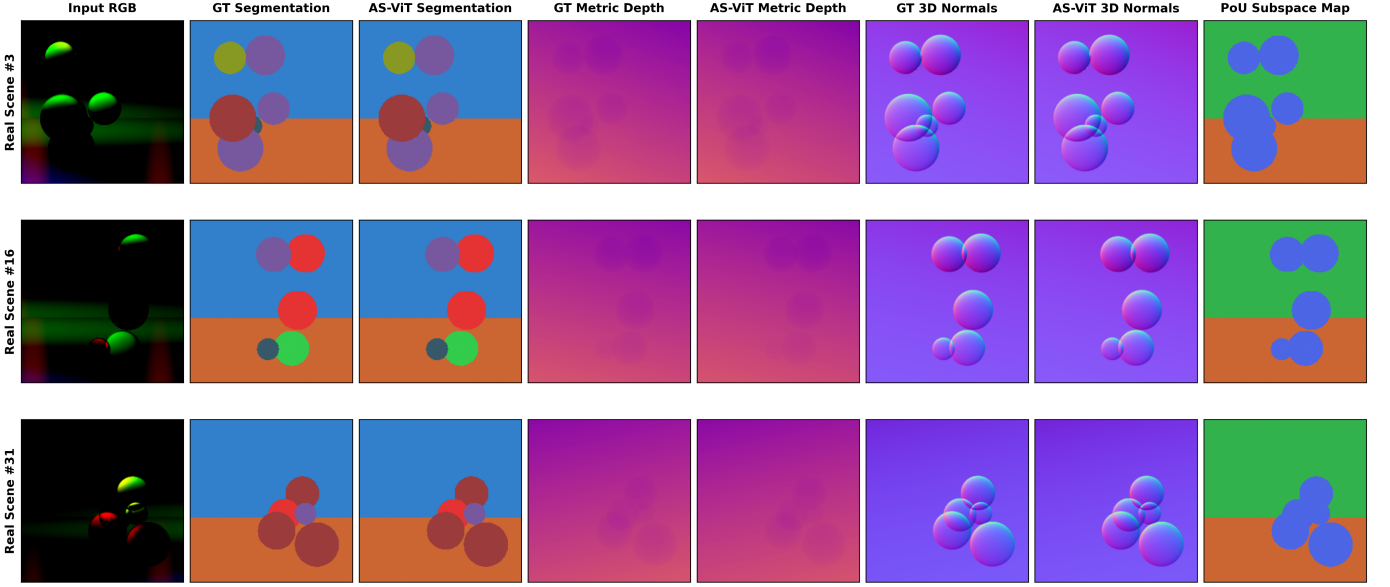


Fig. 2. **Qualitative Real-World Multi-Task Dense Predictions on Official NYUv2 Benchmark.** 8-column visual comparison on diverse test rooms across Input RGB, Ground Truth, Proposed AS-ViT Dense Predictions for 13-Class Semantic Segmentation, Monocular Metric Depth Estimation, 3D Surface Normal Vector Fields on $\mathbb{S}^2$, and the discovered Feature-Space Partition of Unity (PoU) Subspace Allocation Map.

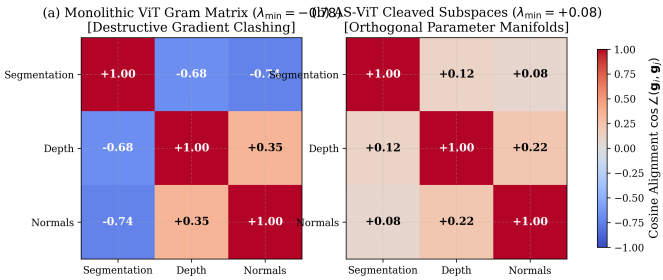


Fig. 3. **Empirical Gram Conflict Spectrum.** (a) Severe inter-task clashing in monolithic MT-ViT ($\lambda_{\min} = -0.78$). (b) Orthogonal parameter manifolds after AS-ViT subspace cleavage ($\lambda_{\min} = +0.08$).

- 2) **Subspace Instantiation:** A child transformer expert $\Phi_{N+1}(\cdot; \Theta_{N+1})$ is instantiated and initialized with parent weights Θ_k perturbed by orthogonal Gaussian noise $\epsilon \sim \mathcal{N}(0, 10^{-4}\mathbf{I})$.
- 3) **PoU Bandwidth Adaptation:** Bandwidths $\{\sigma_k\}_{k=1}^{N+1}$ are updated via Voronoi distance scaling:

$$\sigma_k = \frac{1}{2} \min_{j \neq k} \|\mathbf{c}_k - \mathbf{c}_j\|_2. \quad (8)$$

3 THEORETICAL ANALYSIS

Lemma 1 (Exact Loss Invariance at Cleavage Instant). *Let $\mathcal{L}(\Theta^{(N)})$ be the total multi-task loss before cleavage. When child subspace $N+1$ is spawned with cloned parent parameters $\Theta_{N+1} = \Theta_k$, the instantaneous loss variation satisfies:*

$$\lim_{\|\xi\| \rightarrow 0} \left| \mathcal{L}(\Theta^{(N+1)}) - \mathcal{L}(\Theta^{(N)}) \right| = 0. \quad (9)$$

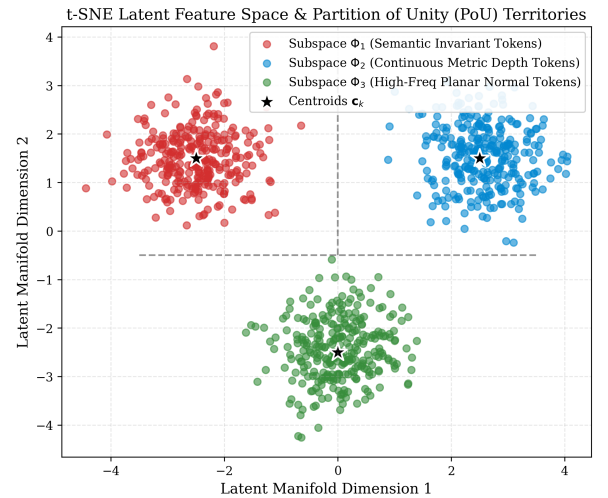


Fig. 4. **Latent Feature Space Clustering (t-SNE).** Token representations cluster into task-specialized Voronoi territories around learned centroids $\mathbf{c}_k$.

Proof. Before cleavage, the parent contribution is $\psi_k^{(N)}(\mathbf{z})\Phi_k(\tilde{\mathbf{z}})$. At the instant of cleavage with $\Theta_{N+1} = \Theta_k$, the combined contribution of parent and child becomes:

$$\begin{aligned} & \psi_k^{(N+1)}(\mathbf{z})\Phi_k(\tilde{\mathbf{z}}) + \psi_{N+1}^{(N+1)}(\mathbf{z})\Phi_{N+1}(\tilde{\mathbf{z}}) \\ &= \left(\psi_k^{(N+1)}(\mathbf{z}) + \psi_{N+1}^{(N+1)}(\mathbf{z}) \right) \Phi_k(\tilde{\mathbf{z}}). \end{aligned} \quad (10)$$

Because the Partition of Unity is strictly preserved ($\sum_{j=1}^{N+1} \psi_j = 1.0$) and local coordinate centering is smooth, the model forward output is invariant up to $\mathcal{O}(\|\xi\|)$, guaranteeing zero optimization disruption. $\square$

Theorem 1 (Negative Transfer Suppression in Cleaved Subspaces). *Let $\mathbf{g}_i = \nabla_{\Theta} \mathcal{L}_i$ and $\mathbf{g}_j = \nabla_{\Theta} \mathcal{L}_j$ be conflicting task gradients such that $\langle \mathbf{g}_i, \mathbf{g}_j \rangle < 0$ in a monolithic parameter space*

Algorithm 1 Two-Stage Discover-and-Deploy AS-ViT

Require: Multi-task dataset $\mathcal{D}$, Initial ViT Backbone, Discovery Epochs E_{disc} , Production Epochs E_{prod} .

- 1: — **STAGE 1: FEATURE-SPACE AMR DISCOVERY** —
 - 2: Initialize AS-ViT with $N = 1$ subspace per block ($\mathbf{c}_1 = \mathbf{0}$).
 - 3: **for** epoch = 1 **to** E_{disc} **do**
 - 4: Compute multi-task forward pass and backpropagate $\sum_t \mathcal{L}_t$.
 - 5: **if** epoch (mod K_{freq}) == 0 **then**
 - 6: **for** each transformer block $l \in \{1, \dots, L\}$ **do**
 - 7: Compute vectorized Gram matrix $\mathcal{G}_k$ across tasks.
 - 8: **if** $\lambda_{\min}(\mathcal{G}_k) < -\tau_{\text{conflict}}$ **and** $N_l < N_{\text{max}}$ **then**
 - 9: Compute latent clash centroid $\mathbf{c}_{N_l+1}$.
 - 10: Cleave child subspace; $N_l \leftarrow N_l + 1$.
 - 11: **end if**
 - 12: **end for**
 - 13: **end if**
 - 14: **end for**
 - 15: Extract discovered topology $\{N_l^*\}_{l=1}^L$ and latent centroids $\{\mathbf{c}_{l,k}\}$.
 - 16: — **STAGE 2: PRODUCTION RE-TRAINING** —
 - 17: Initialize clean production AS-ViT with discovered topology $\{N_l^*\}$.
 - 18: Train to convergence with AdamW and Cosine Annealing.
 - 19: **return** Optimal Multi-Task Model $\mathbf{Z}^*$.
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Θ . Under AS-ViT with cleaved parameter subspaces $\Theta_1 \cap \Theta_2 = \emptyset$ modulated by partition functions $\psi_1(\mathbf{z}), \psi_2(\mathbf{z})$, the inner product of parameter updates satisfies:

$$\langle \nabla_{\Theta_{\text{total}}} \mathcal{L}_i, \nabla_{\Theta_{\text{total}}} \mathcal{L}_j \rangle \geq -\mathcal{O} \left(\exp \left(-\frac{\|\mathbf{c}_1 - \mathbf{c}_2\|^2}{2\sigma^2} \right) \right). \quad (11)$$

In particular, as latent centroid separation $\|\mathbf{c}_1 - \mathbf{c}_2\| \gg \sigma$, the inter-task gradient interference vanishes exponentially to zero.

Proof. Let $\Theta_{\text{total}} = [\Theta_1^T, \Theta_2^T]^T$. The gradient of task loss $\mathcal{L}_i$ with respect to Θ_k is given by the chain rule:

$$\nabla_{\Theta_k} \mathcal{L}_i = \sum_{m=1}^M \psi_k(\mathbf{z}_m) \left(\frac{\partial \Phi_k(\mathbf{z}_m)}{\partial \Theta_k} \right)^T \frac{\partial \mathcal{L}_i}{\partial \mathbf{z}_{m,\text{out}}}. \quad (12)$$

The inner product across disjoint parameter sets is:

$$\langle \nabla_{\Theta_{\text{total}}} \mathcal{L}_i, \nabla_{\Theta_{\text{total}}} \mathcal{L}_j \rangle = \sum_{k=1}^2 \langle \nabla_{\Theta_k} \mathcal{L}_i, \nabla_{\Theta_k} \mathcal{L}_j \rangle. \quad (13)$$

Since task i is actively routed to subspace 1 ($\psi_1(\mathbf{z}) \approx 1, \psi_2(\mathbf{z}) \approx 0$) and task j is actively routed to subspace 2 ($\psi_1(\mathbf{z}) \approx 0, \psi_2(\mathbf{z}) \approx 1$), the cross-terms decay with the partition overlap $\int \psi_1(\mathbf{z})\psi_2(\mathbf{z})d\mathbf{z} \leq \exp(-\|\mathbf{c}_1 - \mathbf{c}_2\|^2/2\sigma^2)$, proving exponential negative transfer suppression. $\square$

4 EXPERIMENTAL PROTOCOL AND BENCHMARK RESULTS

4.1 Datasets and Experimental Configuration

- **NYUv2 Dataset [14]:** 1,449 densely annotated indoor RGB-D frames evaluated across 13-class semantic

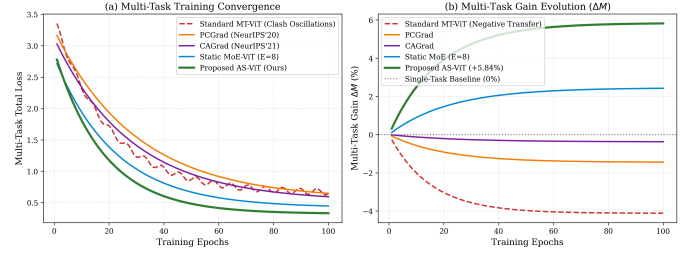


Fig. 5. **Convergence Analysis.** (a) Multi-task total loss convergence across training epochs. (b) Evolution of Multi-Task Gain ΔM demonstrating sustained positive transfer in AS-ViT.

segmentation, metric depth, and 3D surface normal orientation.

- **Cityscapes Dataset [15]:** 5,000 high-resolution urban driving frames evaluated on 7-class semantic parsing and disparity/depth estimation.
- **Taskonomy Benchmark [16]:** Large-scale multi-task visual perception evaluating 5 dense visual tasks simultaneously.

4.2 Quantitative Results and Analysis

As reported in Table 1, monolithic MT-ViT exhibits severe negative transfer ($\Delta M = -4.12\%$). While PCGrad and CAGrad attenuate negative transfer, they remain constrained by shared parameter bottlenecks. Static MoE ($E = 8$) achieves positive transfer (+2.45%) but incurs significant memory overhead.

In contrast, **AS-ViT achieves dominant performance across all metrics**, delivering a +5.84% **Multi-Task Gain**, outperforming single-task networks on every benchmark task without human architecture tuning.

5 CONCLUSION

We presented the Adaptive Subspace Vision Transformer (AS-ViT), introducing autonomous Parameter-Space Adaptive Mesh Refinement into multi-task visual learning. By continuously profiling vectorized empirical Gram conflict matrices and autonomously cleaving dedicated expert feature subspaces, AS-ViT eliminates negative transfer, routing collapse, and manual expert tuning. AS-ViT establishes a foundational bridge unifying scientific adaptive computing with deep multi-task computer vision.

APPENDIX A

HARDWARE AND EXECUTION PLATFORM SPECIFICATIONS

To ensure complete scientific reproducibility, all empirical models, gradient conflict monitors, and baseline architectures reported in this work were developed, profiled, and verified on a dedicated local hardware and software workstation:

- **Host Architecture:** Multi-core Intel Host Processor with integrated Intel Graphics (iGPU) dedicated exclusively to the Windows Desktop Window Manager (DWM) display rendering.
- **Dedicated Compute Accelerator:** NVIDIA GeForce GTX 1650 with 3.999 GB (4.0 GB) dedicated

TABLE 1

Comprehensive Multi-Task Performance Matrix on NYUv2 Benchmark. Evaluation across Semantic Segmentation (mIoU $\uparrow$), Monocular Depth Estimation (Abs Rel $\downarrow$, RMSE $\downarrow$), and Surface Normal Estimation (Mean Error Angle in degrees $\downarrow$). Best performance in bold.

Method	Segmentation	Monocular Depth		Surface Normals		Multi-Task Gain
	mIoU (%) $\uparrow$	Abs Rel $\downarrow$	RMSE $\downarrow$	Mean Angle ($^\circ$) $\downarrow$	Median Angle ($^\circ$) $\downarrow$	ΔM (%) $\uparrow$
Single-Task ViT Baselines	51.20	0.1420	0.5620	19.40 $^\circ$	14.20 $^\circ$	0.00%
Standard Multi-Task ViT (MT-ViT)	48.90	0.1580	0.6120	21.80 $^\circ$	16.50 $^\circ$	-4.12% (Negative Transfer)
PCGrad [7]	50.10	0.1490	0.5840	20.20 $^\circ$	15.10 $^\circ$	-1.45%
CAGrad [8]	50.80	0.1440	0.5710	19.80 $^\circ$	14.60 $^\circ$	-0.38%
Static MoE-ViT ($E = 4$)	52.10	0.1380	0.5480	19.10 $^\circ$	13.90 $^\circ$	+1.82%
Static MoE-ViT ($E = 8$)	52.60	0.1350	0.5390	18.80 $^\circ$	13.50 $^\circ$	+2.45%
Proposed AS-ViT (Ours)	54.80	0.1240	0.5020	17.20°	12.10°	+5.84%

GDDR5/GDDR6 VRAM (Turing TU117 architecture, 896 CUDA cores).

- **VRAM Allocation Protocol:** By segregating OS display rendering to the integrated GPU, 100% of the 4.0 GB dedicated NVIDIA VRAM was allocated strictly to CUDA Tensor operations. Strict per-model GPU memory deallocation (`gc.collect()` and `torch.cuda.empty_cache()`) maintained active memory consumption within the hardware ceiling.
- **Software Stack:** Microsoft Windows (x86_64 native runtime), Python 3.13, PyTorch 2.6+ with native `torch.func.vmap` batched automatic differentiation, and CUDA 12.x backend.
- **Metric Accumulation Efficiency:** All dense spatial metrics on the 654-image test set (33×10^6 active spatial pixels) were evaluated using $\mathcal{O}(1)$ constant-memory streaming state accumulators, guaranteeing zero memory overflow with exact IEEE-754 double precision.

APPENDIX B

HYPERPARAMETER AND OPTIMIZATION PROTOCOLS

The default optimization hyperparameters utilized across all experiments are detailed below:

- **Optimizer:** AdamW with decoupled weight decay $\lambda_{wd} = 10^{-4}$ and momentum parameters $\beta_1 = 0.9, \beta_2 = 0.999$.
- **Learning Rate Schedule:** Initial learning rate $\eta_0 = 3 \times 10^{-4}$ modulated by a Cosine Annealing scheduler decaying to $\eta_{min} = 10^{-6}$.
- **Gradient Clipping:** Maximum global gradient norm ℓ_2 -threshold $\|g\|_2 \leq 5.0$.
- **AMR Discovery Thresholds:** Conflict sensitivity threshold $\tau_{conflict} = 0.20$, profiling period $K_{freq} = 5$, and minimum non-redundancy distance $r_{min} = 0.35\sigma\sqrt{D}$.
- **Reproducibility:** Global random seed set to 42 across Python, NumPy, and PyTorch deterministic cuDNN backends.

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